HARD · ADVANCED MATH

Equivalent expressions

01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 ADVANCED MATH

The equation $\frac{x^2+6x-7}{x+7}=ax+d$ is true for all $x\neq-7$, where a and d are integers. What is the value of $a+d$?

A. -6

B. -1

C. 0

D. 1

Q2 GRID-IN ADVANCED MATH

$$\sqrt[5]{70n}\ \sqrt[6]{70n}\ ^{2}$$

For what value of x is the given expression equivalent to $70n^{30x}$, where $n > 1$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q3

ADVANCED MATH

Which expression is equivalent to $\frac{y + 12}{x - 8} + \frac{yx - 8}{x^2y - 8xy}$?

A. $\frac{xy + y + 4}{x^3y - 16x^2y + 64xy}$

B. $\frac{xy + 9y + 12}{x^2y - 8xy + x - 8}$

C. $\frac{xy^2 + 13xy - 8y}{x^2y - 8xy}$

D. $\frac{xy^2 + 13xy - 8y}{x^3y - 16x^2y + 64xy}$

Q4

ADVANCED MATH

$$f(x) = x^3 - 9x$$
$$g(x) = x^2 - 2x - 3$$

Which of the following expressions is equivalent to $\dfrac{f(x)}{g(x)}$, for $x > 3$?

A. $\dfrac{1}{x+1}$

B. $\dfrac{x+3}{x+1}$

C. $\dfrac{x(x-3)}{x+1}$

D. $\dfrac{x(x+3)}{x+1}$

The expression $\frac{x^{-2}y^{\frac{1}{2}}}{x^{\frac{1}{3}}y^{-1}}$, where $x > 1$ and $y > 1$, is equivalent to which of the following?

A. $\frac{\sqrt{y}}{\sqrt[3]{x^2}}$

B. $\frac{y\sqrt{y}}{\sqrt[3]{x^2}}$

C. $\frac{y\sqrt{y}}{x\sqrt{x}}$

D. $\frac{y\sqrt{y}}{x^2\sqrt[3]{x}}$

Q6

ADVANCED MATH

$$\frac{2}{x-2}+\frac{3}{x+5}=\frac{rx+t}{(x-2)(x+5)}$$

The equation above is true for all $x>2$, where r and t are positive constants. What is the value of rt ?

A. -20

B. 15

C. 20

D. 60

Q7 ADVANCED MATH

$$(ax+3)(5x^2-bx+4)=20x^3-9x^2-2x+12$$

The equation above is true for all x, where a and b are constants. What is the value of ab ?

A. 18

B. 20

C. 24

D. 40

Q8 GRID-IN ADVANCED MATH

If $\frac{\sqrt{x^5}}{\sqrt[3]{x^4}} = x^{\frac{a}{b}}$ for all positive values of x, what is the value of $\frac{a}{b}$?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q9

GRID-IN ADVANCED MATH

One of the factors of $2x^3 + 42x^2 + 208x$ is $x + b$, where b is a positive constant. What is the smallest possible value of b?

STUDENT-PRODUCED RESPONSE

.	/ .	/ .	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q10 ADVANCED MATH

Which expression is equivalent to $\frac{4}{4x - 5} - \frac{1}{x + 1}$?

A. $\frac{1}{x + 14x - 5}$

B. $\frac{3}{3x - 6}$

C. $-\frac{1}{x + 14x - 5}$

D. $\frac{9}{x + 14x - 5}$